

DATA 505 (Statistics Using R): Exam 2

Part 2: Open Computer

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Instructions

Welcome to Part 2 of the exam!

You now have access to your computer and any online resources that are “not alive” (i.e., you may freely use online resources, but you may not communicate with any other person). You have approximately 45 minutes to complete this part.

Fill in the code chunks below and provide written answers where requested. Submit this completed .qmd file on Moodle along with the rendered PDF.

Setup and Data Loading

We will analyze the `MA_schools` dataset from the `moderndive` package, which contains data on Massachusetts public high schools in 2017.

```
# Install moderndive package if not already installed
if (!require(moderndive)) {
  install.packages("moderndive")
}

# Load required packages
library(moderndive)
library(ggplot2)

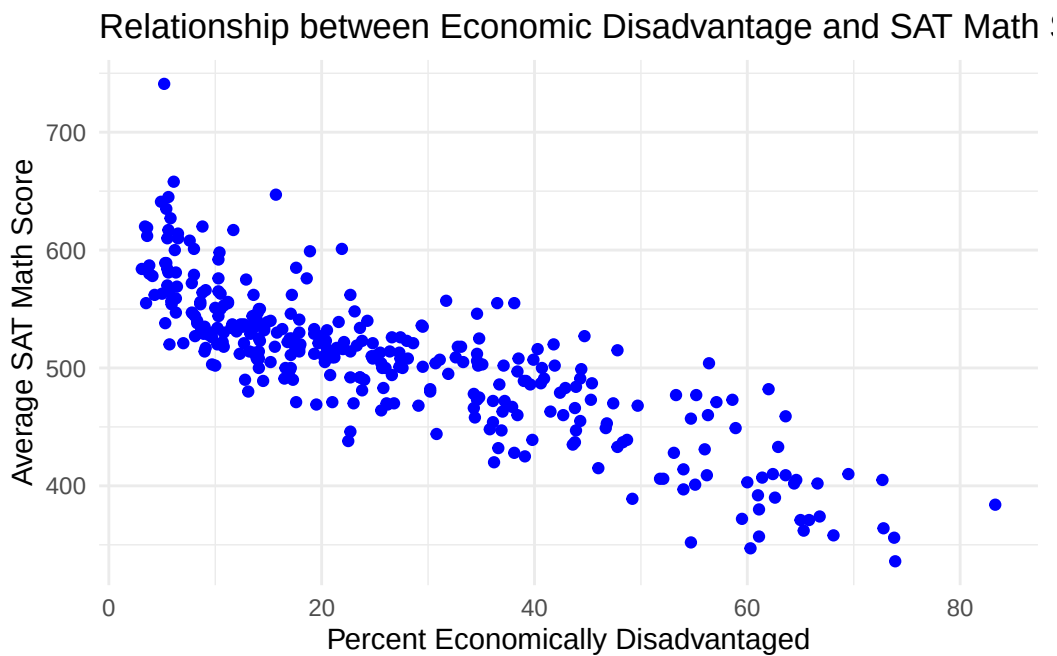
# Load the MA_schools dataset
data(MA_schools)
```

Data Exploration (3 points)

1 (3 points)

Create a scatterplot showing the relationship between `perc_disadvan` (percent economically disadvantaged students) and `average_sat_math` (average SAT math score). Based on the plot, do you expect a positive or negative relationship?

```
# YOUR CODE HERE
ggplot(MA_schools, aes(x = perc_disadvan, y = average_sat_math)) +
  geom_point(color = "blue") +
  labs(
    x = "Percent Economically Disadvantaged",
    y = "Average SAT Math Score",
    title = "Relationship between Economic Disadvantage and SAT Math Scores"
  ) +
  theme_minimal()
```



Answer: WRITE YOUR ANSWER HERE. There is a clear negative relationship — as the percentage of economically disadvantaged students increases, the average SAT math score decreases.

Simple Linear Regression (12 points)

1 (4 points)

Fit a simple linear regression model predicting `average_sat_math` from `perc_disadvan`. Store the model as `model1` and display the summary.

```
# YOUR CODE HERE
model1 <- lm(average_sat_math ~ perc_disadvan, data = MA_schools)
summary(model1)
```

Call:

```
lm(formula = average_sat_math ~ perc_disadvan, data = MA_schools)
```

Residuals:

Min	1Q	Median	3Q	Max
-80.74	-21.26	-4.12	18.54	174.17

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	581.2811	3.2668	177.9	<2e-16 ***
perc_disadvan	-2.7798	0.1011	-27.5	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.54 on 330 degrees of freedom

Multiple R-squared: 0.6962, Adjusted R-squared: 0.6953

F-statistic: 756.2 on 1 and 330 DF, p-value: < 2.2e-16

2 (4 points)

Extract the estimated model coefficients (intercept and slope) from this fitted model object, and store them in a vector called `sat_effects`. Then report the intercept and slope below.

```
# YOUR CODE HERE
sat_effects <- coef(model1)
sat_effects
```

(Intercept)	perc_disadvan
581.281054	-2.779787

Intercept: WRITE YOUR ANSWER HERE. 581.28

Slope: WRITE YOUR ANSWER HERE. -2.78 ### (4 points)

Interpret the slope coefficient in context. What does it tell you about the relationship between the percentage of economically disadvantaged students and average SAT math scores?

Interpretation: WRITE YOUR ANSWER HERE. For every 1 percentage point increase in the percentage of students facing economic hardship, the school's average SAT math score is projected to decrease by approximately 2.78 points, indicating a significant negative correlation between the two. ## Making Predictions (8 points)

3 (4 points)

Use `model1` to predict the average SAT math score for a school where 50% of students are economically disadvantaged. Create a data frame with this value and use the `predict()` function.

```
# YOUR CODE HERE
new_school <- data.frame(perc_disadvan = 50)
predict(model1, newdata = new_school)
```

```
      1
1 442.2917
```

Predicted average SAT math score: WRITE YOUR ANSWER HERE. When 50% of students are from economically disadvantaged backgrounds, the school's average SAT Math score is projected to be approximately 442.29. ### (4 points)

Calculate and print out a 95% prediction interval for a single new school where 50% of students are economically disadvantaged.

```
# YOUR CODE HERE
predict(model1, newdata = new_school, interval = "prediction", level = 0.95)
```

```
      fit      lwr      upr
1 442.2917 376.0542 508.5292
```

Multiple Linear Regression (12 points)

1 (4 points)

Fit a multiple regression model predicting `average_sat_math` from both `perc_disadvan` and `size` (school size category). Store this as `model2` and display the summary.

```
# YOUR CODE HERE
model2 <- lm(average_sat_math ~ perc_disadvan + size, data = MA_schools)
summary(model2)
```

Call:

```
lm(formula = average_sat_math ~ perc_disadvan + size, data = MA_schools)
```

Residuals:

Min	1Q	Median	3Q	Max
-81.342	-20.802	-3.872	18.211	167.251

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	588.1898	7.6067	77.325	<2e-16 ***
perc_disadvan	-2.7772	0.1063	-26.120	<2e-16 ***
sizemedium	-11.9134	7.5346	-1.581	0.115
sizelarge	-6.3615	6.9231	-0.919	0.359

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 33.5 on 328 degrees of freedom

Multiple R-squared: 0.6988, Adjusted R-squared: 0.696

F-statistic: 253.6 on 3 and 328 DF, p-value: < 2.2e-16

2 (4 points)

Compare the R-squared values of `model1` and `model2`. Which model explains more variance in average SAT math scores?

```
# YOUR CODE HERE
summary(model1)$r.squared
```

```
[1] 0.6961948
```

```
summary(model2)$r.squared
```

```
[1] 0.6987587
```

Answer: WRITE YOUR ANSWER HERE. We can see that the R^2 of model 1 is 0.696, and the R^2 of model 2 is 0.699. Therefore, model 2 explains slightly more variance, but the improvement is very small. ### (4 points)

Use the `anova()` function to formally compare `model1` and `model2`. Does adding `size` significantly improve the model at the $\alpha = 0.05$ level?

```
# YOUR CODE HERE
anova(model1, model2)
```

Analysis of Variance Table

```
Model 1: average_sat_math ~ perc_disadvan
Model 2: average_sat_math ~ perc_disadvan + size
  Res.Df    RSS Df Sum of Sq    F Pr(>F)
1     330 371191
2     328 368058  2     3132.6 1.3958 0.2491
```

Conclusion: WRITE YOUR ANSWER HERE. Since the p-value is $0.2491 > 0.05$, adding the variable 'size' did not significantly improve the explanatory power of the model at a significance level of $\alpha = 0.05$. Therefore, I believe that model 1 is sufficient to explain the variation in SAT math scores.